

Mechanics

This section will include detailed numbers on various mechanics within the game. Often, you don't need to know them in such detail.

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Gacha

The SSR/3* rates in Umamusume are 3%, with rate-up cards and umas appearing 0.75% of the time usually. You'll get the rate-up thing $\frac{1}{4}$ times you get an SSR/3*. This is the same whether there are two rate-ups or only one. Each rate-up will be 0.75%, it doesn't go up to 1.5% for single rate-ups. Because of this, dual banners tend to be preferred to pull on.

You also get to pick any of the rate-up things after doing 200 pulls, which is known as sparking, a term originating from Granblue Fantasy. (though, it's a bit confusing with inheritance also being sparks on Global...) To get MLB SSRs, you will on average need to spark twice. Pities do not carry over across banners. If you do 100 pulls on one banner, you will start from 0 on the next, not from 100.

In terms of USD, if you want to whale, one spark will run you \$240 if you get them on sale, and about \$408 at full price, or roughly two dollars per pull. The 2500 pack is the "best value" after getting all the sales and monthly pass. They renew the sale packs from time to time.

There are no limited umas. You can pull any character, even the summer or Christmas ones, at any time. Some welfare support cards are event-exclusive, but otherwise there are also no limited support cards.

Here's a table showing what you can expect from certain numbers of pulls, pity included, for any given rate-up.

SSR

Pulls	None	0LB	1LB	2LB	3LB	MLB
50	69%	26%	5%	0.6%	0.1%	0%
100	47%	36%	13%	3%	0.6%	0.1%
200	0%	22%	34%	25%	13%	6%
300	0%	11%	24%	27%	20%	19%
400	0%	0%	5%	15%	22%	58%

500	0%	0%	2%	9%	17%	72%
600	0%	0%	0%	1%	5%	94%

Sometimes, there will be an uma banner with no rate up. This happened for New Year's 2021 on the JP server. In this case, you can spark any uma.

Legacies

3 Star Spark Chance

After finishing a run, the uma gets random sparks to pass down to its future children. The blue one is chosen randomly between the five stats, and then the stars are chosen based on how high that stat is. The pink is chosen randomly between all aptitudes the uma has at A or higher (so, avoid raising more to A if you're targeting a specific pink). Here's a chart of the chance to get a 3 star spark based on what the stat is.

Stat Value	<600	600~1100	>1100
1 Star Factor	~90%	~50%	~20%
2 Star Factor	~10%	~45%	~70%
3 Star Factor	0%	~6%	~10%

After finishing an uma, a stat is randomly chosen, and then the stars for that stat are chosen. So, if you had 1200 Speed, the chance of getting a 3 star Speed spark would be 20% (the chance of it picking Speed) * 10% (the chance of it being 3 star) for a total of ~2%. It doesn't matter if you win the run or not.

Raising Aptitude

During the initial inheritance, the amount the aptitude increases is based on the number of total stars among the pink sparks of the parents and grandparents. Raising it by one rank requires 1 star, then each subsequent rank requires another three stars. You can raise each aptitude a total of four ranks in this way, so an F rank aptitude could only go to B during the initial inheritance, even with 18 total stars.

As an example, Mayano Top Gun has an E rank Dirt aptitude and a D rank Mile aptitude. In order to raise her for Dirt PvP, you would need to raise both to A. Getting Mile to C takes one star, then 3 to get to B, and 3 more to get to A, for a total of 7. The Dirt aptitude requires one more step, so it requires 10 total stars.

In total, this is 17, but it's not possible to do all at the start since one uma can't have two different pink aptitudes. You could do 7 Mile and 9 Dirt ($3+3+1$, $3+3+3$), or 6 Mile and 10 Dirt ($3+3$, $1+3+3+3$). One of them will have to be left at B and raised through an inspiration event.

Chance of Inheriting Sparks

The chance of inheriting sparks is influenced by your individual affinity with each uma and whether the uma is a parent or grandparent. Here's a table I stole from [Crazyfellow](#).

Spark type↓ / Base chance→	☆1 Rate	☆2 Rate	☆3 Rate
Blue	70%	80%	90%
Pink	1%	3%	5%
Green (Unique Skill)	5%	10%	15%
Race Whites	1%	2%	3%
Other Whites	3%	6%	9%

These numbers are then increased by a % equalling your affinity. So, if you had 30 affinity with your parent that had a 3 star skill white, you would have a $9\% * 1.3 = \sim 12\%$ chance of inheriting it instead of 9%.

Chance of Getting White Sparks

When you finish a run, you have a chance to get white sparks for each skill you have. If it's a white skill, it's a 20% chance. If it's a © skill, it's 25%. If it's a gold skill, it's 40%. For each parent or grandparent that also has the spark, you add 2.5%, or 5% for the gold skills.

Other white sparks, such as race and scenario sparks, follow the same pattern, counting as white skills (so 20% base chance).

For stars, it's 50% for 1 star, 45% for 2 stars, and 5% for 3 stars. These change to 20% for 1 star, 70% for 2 star, and 10% for 3 star **if the uma is SS rank or higher**.

Affinity

The better the affinity, the higher the chance of getting a lot of inherited sparks during the two inspiration events during the training, which happen in early April Y2 & Y3.



The affinity is decided by a variety of things. At a baseline, each uma has hard-coded affinity with every other uma, which is derived from various things usually related to the real-life umas. Dirt racers tend to have good affinity with other Dirt racers and such. This affinity applies between the uma you're training and her parents, and between her and her grandparents. Every uma has 0 affinity with themselves, which is significant for grandparents. You can check these numbers on [Gametora](#).

Triangle is 50 or less affinity, double circle is greater than 150 affinity. There's no significant jump here, 149 affinity isn't massively worse than 151. Affinity is linearly useful, 149->150 is the same increase as 150->151. But, the double circle is a good target when you don't want to bother calculating everything.

You can increase this affinity by doing races. Currently, you get +1 affinity for each graded race you win on both the parent and the grandparent. So, if you won 10 graded races on Oguri, then trained a Teio with the Oguri as a parent and did those same 10 races, you would get +10 affinity for the Teio parent, making it easier to get double circle affinity for your runs. If the other grandparent also did those 10 races, you would have +20 affinity.

If both parents did the race, there's another bonus. So, if the other parent was a Mayano who did those 10 races, it would be another +10 affinity, for +30 total. Doing races is also good since it gives you more SP, which means more skills, which means more chances for white sparks.

However, later on this gets changed to be +3 per matching G1, with G2s and G3s no longer counting. This applies retroactively, so your parents' affinities might change after that update. As such, just focus on the G1s when making your parents in this way. They [give good stats](#) anyway, you should be doing them with some frequency even on normal runs

Stats

Speed is always an important stat. You will want as much of it as possible, as it increases your top speed during the last spurt.

BaseSpeed = $(20 - (\text{CourseLength} - 2000) / 1000) * \text{PhaseModifier}$

FinalLegSpeed = $\text{BaseSpeed} + \sqrt{500 * \text{Speed}} * \text{DistanceAptitude} * 0.002$

MaxSpurtSpeed = $(\text{FinalLegSpeed} + 0.01 * \text{BaseSpeed}) * 1.05 + \sqrt{500 * \text{Speed}} * \text{DistanceAptitude} * 0.002$

Stamina is necessary up to a point, and then starts to fall off depending on distance. How much you need is based on a variety of factors. The lower your speed, the less you'll need. [Pace Chasers also receive an 11% Stamina penalty.](#) For a target stamina for each style in each distance, see the [Required Stamina Chart](#) below.

Long distance has a wide variety of distances, from 2500m up to 3600m. If you're targeting ones on the shorter end, you can skimp more on stamina. They're 2600m or lower over half of the time, with the other half being 3000+m in Team Trials.

On top of causing you to lose Speed when you run out of HP, your remaining HP also affects how long and how fast your last spurt will be. If you lack Stamina, your uma will start her last spurt later or run slower to use her HP more efficiently. If you completely run out of HP, you'll quickly decelerate.

Power is more important in shorter distances, but always helps out. It affects how quickly you accelerate to your top speed. In longer races, you have more time to accelerate, so having a high acceleration doesn't have as big of an impact. For the most part, if you meet your stamina goal, you'll focus as much on speed as possible, then power second.

BaseAcceleration = 0.0006 normally, 0.0004 on uphill.

Acceleration = BaseAcceleration * $\sqrt{500 * \text{Power}}$ * [PhaseModifier](#) * [SurfaceAptitude](#)

Power also affects your speed on uphill. When going uphill, you receive a speed penalty of [\[100 * tan\(angle in radians\)\] * 200 / Power](#).

It also affects how quickly you change lanes. Your lane change speed is $0.02 * (0.3 + 0.001 * \text{Power})$ m/s.

Guts isn't useful in large numbers, but you need a small amount of it. Umas will want to have ~300 guts, but rarely more than 400. Up to 400, it's close to equivalent to stamina, while after, it falls off. See the [Stamina vs Guts](#) section for details.

The Final Leg HP consumption modifier is [1.0 + 200/sqrt\(600 * Guts\)](#).

Wit, aka Wisdom affects your skill trigger rates, as well as some other things. Aim to have between 300 and 500 wisdom for any uma. Any more than that and it starts to fall off too much. You can see a chart of the skill proc chance based on your wisdom [here](#).

Wisdom	200	300	400	500	600	700
Skill Proc	55%	70%	77.5%	82%	85%	87%

Wit also affects the chance that the uma suffers from “**Rushed**” debuff. Whether the uma will suffer Rushed is determined at the start of the race based on a Wisdom check. The chance is $(6.5 / \log_{10}(0.1 * Wit + 1))^2$ percent.

Rushed will occur at a mostly random point in the first half of the course. It lasts for 12 seconds, but you have a 55% chance to break free every 3 seconds. While under Rushed, the uma will consume HP 60% faster and switch their style to a more aggressive one (eg, a Late Surger might act as a Pace Chaser or a Front Runner, Front Runner acts like Twin Turbo).

Wisdom	200	300	400	500	600	700
Rushed	24.17%	19%	16.24%	14.49%	13.26%	12.32%

Wit can also give you a chance to increase your speed on downhills. The speed boost you get from downhills is $0.3 + |100 * \tan(\text{angle in radians})| / 10$, and the chance you receive it is $0.04\% * \text{Wisdom}$. So, 4% per 100 Wit. Every second, this has a 20% chance to end.

Another other thing Wit does is help determine your target speed at each section of the race. There are 24 such sections, and in each of them, your uma applies a random modifier to her target speed, with the help of your Wit.

The max modifier is $Wit / 5500 * \log_{10}(Wit * 0.1)\%$

The minimum modifier is the max speed minus 0.65.

Wisdom	200	300	400	500	600	700
Max Mod	0.047%	0.081%	0.117%	0.154%	0.194%	0.235%
Min Mod	-0.603%	-0.569%	-0.533%	-0.496%	-0.456%	-0.415%

Stat Importance

Assuming a perfect uma with 1200 in all stats, but no skills, here is how much faster that uma is than an uma with 1200 in all stats except one, measured in lengths. The “Long” track is Tenno Sho Spring, the “Medium” track is Derby, the “Sprint” is Sprinter’s Stakes.

This table was made before the update that made Guts affect spurt speed.

Stat	400	600	800	1000
Speed (Long)	21.29	14.48 (+6.81)	8.92 (+5.56)	4.18 (+4.74)
Speed (Sprint)	5.04	3.31 (+1.73)	1.98 (+1.33)	0.90 (+1.08)
Stamina (Long)	45.07	42.17 (+2.90)	25.50 (+16.67)	10.46 (+15.04)
Stamina (Med)	21.35	6.48 (+14.87)	0.00 (+6.48)	0.00
Power (Long)	19.64	10.03 (+9.61)	5.12 (+4.91)	2.10 (+3.02)
Power (Sprint)	7.92	4.61 (+3.31)	2.53 (+2.08)	1.10 (+1.43)
Guts (Long)	19.38	10.05 (+9.33)	4.27 (+5.78)	1.48 (+2.79)
Guts (Med)	0.00	0.00	0.00	0.00
Wit (Long)	5.58	4.30 (+1.28)	2.93 (+1.37)	1.49 (+1.44)

Acceleration makes Power less important. For example, if the Sprint uma is Pasa, then Speed outperforms Power. And, training Speed raises Power, while training Power raises Stamina, so even in Sprint, Speed wins out in the end. Power is also stronger the more Speed you have, as the higher your top speed, the more you need to accelerate. And vice versa, since having higher Power lets you accelerate to your top speed faster, getting more out of your Speed stat.

Stamina and Guts are extremely powerful when giving you more effective HP, but useless otherwise, with Guts being a spicy version of Stamina. (see next page) Wit also gets better when you lack Stamina, as it makes your spurts better and can reduce HP consumption on downhills, acting as bonus Stamina.

From this, we can think of the stat importance as:

Stamina/Guts up to the [track's requirement](#) > Speed > Power > Wit

Stamina vs Guts

<https://twitter.com/ottoh1112/status/1389344141172035585>

Each distance has a “target” Guts. Below this number, each point of Guts lost is worth more than losing a point of Stamina. Above, each point of Guts gained is worth less than gaining a point of Stamina.

In Sprint, this threshold is at 210 Guts.

In Mile and Dirt, 260 Guts.

In Medium, 320 Guts.

In Long, 3000m has a threshold of 380 Guts, while 3600m has one of 440.

Guts is only slightly worse than Stamina while going 50 over this threshold point. Those 50 points of Guts will be worth 43~46 Stamina. The next 50 will be worth 33~37 Stamina. The 50 after that is only worth 30~31.

On the other hand, Guts is better than Stamina when below the threshold. If you're 50 below, then adding 50 Guts is like adding 55~62 Stamina. If you're 100 below, adding 50 is like adding 66~83. At 150 below, the 50 Guts would be worth 82~106 Stamina. It's more impactful in the shorter distances.

HP

The HP calculation formula is $(\text{Track Distance}) + (\text{Stamina} * 0.8) * (\text{Style Modifier})$.

Front Runners have a multiplier of 0.95, or 95%, while Pace Chasers have one of 0.89, Late Surgers have one of 1, and End Closers have one of 0.995.

You might ask how Guts fits into this. Guts lowers your HP consumption at the end of the race. The multiplier for HP consumed is $1 + (200 / \sqrt{600 * \text{Guts}})$. More Guts means a lower HP consumption. You can see a graph of this [here](#).

The base HP consumption rate is $20 * (\text{CurrentSpeed} - \text{BaseSpeed} + 12)^2 / 144$

The track condition also affects this. The worst condition will give a 1.02 multiplier to HP consumed throughout the race, while the second worst gives 1.02 on Turf and 1.01 on Dirt.

While in **Rushed**, HP consumed is multiplied by 1.6. When you trigger the downhill boost, it's multiplied by 0.4. When the uma enters [pace down](#) mode, it's multiplied by 0.6. Front umas never enter pace down.

Recovery skills heal a percentage of your HP. A golden one will restore 5.5%, while a white will restore 1.5%. For Unique skills, if the uma is below 3 stars, it will restore 3.5%, otherwise, 5.5%.

Stamina Calculator

To determine whether your uma has enough stamina, use [this spreadsheet](#), which isn't track-specific but works fine. Most notably, it doesn't consider hills, so will often overestimate your required stamina. You'll want to leave the "consider skill proc rate" option off. Average doesn't make sense for heals, you get them or you don't, so the "[chance of failure](#)" is the important stat rather than your average.

You could also plug their information into [this calculator](#). The options, despite looking like they're all off, actually default to on, so you'll have to turn them on and back off again to get closer to Global. There are probably various differences in the code, from updated mechanics (like Guts) to updated course data, so it will be more accurate the more time passes.

Or, you can use the Umalator, which is sometimes a bit more work. There's a section on using it to check stamina [here](#). All three will usually give results within 100-200 Stamina of each other.

Required Stamina Chart

These numbers are generated with the [umalator](#), for quick reference. Downhills reduce the Stamina requirement, so some tracks will also be lower than this. Eg, Hanshin 3000m takes 150 less stamina than Kyoto 3000m, due to having 1000m of downhills.

The numbers seem extreme, but **Stamina is more important than Speed** if you lack Stamina. 1200 Speed 500 Stamina is **3 seconds (~30 lengths) slower** than 500 Speed 1200 Stamina on Tennosho Spring (3200m). You lose so much speed when out of stamina. In Career Mode, the invisible +400 All Stats bonus makes these much lower. See [Stamina vs Speed](#).

These numbers are calculated with 1200 Speed, 1200 Power, and 600 Wit, on tracks with no downhills, and with no skills other than heals. For Sprint/Mile, it uses 300 Guts, and for Med/Long, 400 Guts. See [Stamina vs Guts](#). The existence of downhills can lower these numbers, and the existence of Debuffers can raise them. Skills can also cause you to spend more stamina due to running faster, but it tends to be minor, like 10-30 Stamina. Front Runner accounts for fighting to keep first place (Spot Struggle/Lead Comp), while Pace Chasers can be content with their positioning and enter pace down, which saves them stamina. So Front Runners use more stamina, even though Pace Chasers have the least.

Required Stamina	Front Runner	Pace Chaser	Late Surger	End Closer
Short - 1400m	570	540	500	510
Mile - 1800m	800	770	720	740

Mile - 1800m + 1 Gold	640	600	560	580
Mid - 2400m + 1 Gold	910	930	870	900
Mid - 2400m + 2 Gold	710	720	680	700
Long - 2600m + 1 Gold	1130	1110	1030	1060
Long - 2600m + 2 Gold	900	870	820	850
Long - 3200m + 2 Gold	1080	1060	990	1020
Long - 3200m + 3 Gold	830	800	750	780

Note that the more gold recoveries you rely on, the less consistent you'll be, as each one only has a roughly 70-80% chance to activate. Check out the chart below to judge how often your uma might end up failing due to missed skill activations.

Recovery Consistency

Required Stamina recommendations will list the required stamina based on the number of recoveries, but it assumes those recoveries activate. Here's the chance of that happening based on your Wit stat and how many recoveries you have.

Wit:	300	400	500	600	700	800
1 Gold Recovery	70.0%	77.5%	82.0%	85.0%	87.1%	88.8%
1 of 2 Gold Recoveries	91.0%	94.9%	96.8%	97.8%	98.3%	98.7%
2 Gold Recoveries	49.0%	60.1%	67.2%	72.3%	75.9%	78.9%
2 of 3 Gold Recoveries	78.4%	87.1%	91.4%	93.9%	95.4%	96.5%
2 of 4 Gold Recoveries	91.6%	96.2%	98.0%	98.8%	99.2%	99.5%
3 Gold Recoveries	34.4%	46.5%	55.1%	61.4%	66.1%	70.0%
3 of 4 Gold Recoveries	65.2%	78.0%	84.9%	89.0%	91.7%	93.6%
4 Gold Recoveries	24.0%	36.1%	45.2%	52.2%	57.6%	62.2%
4 of 5 Gold Recoveries	52.8%	68.5%	77.8%	83.5%	87.3%	90.0%

Note that some recoveries are also inherently unreliable. For example, it's very rare for Iron Will to trigger, Breath of Fresh Air can trigger too early or late to do anything, and Adrenaline Rush will always be too late. See [Gold Recovery Rankings](#).

Stamina vs Speed

So, how important is reaching the Stamina requirement, *really*? Surely skimping on Stamina and increasing Speed will make the uma run faster, *right*? How bad could it be?

Well, here are some comparisons. They assume both umas have 1200 Power, [250 Guts](#), and 1000 Wit, and are verified over 100 1v1 races. The umas have no skills, other than recoveries when mentioned, and their uniques do not trigger. They race on tracks with no downhill, and are both Pace Chasers. There is one Front Runner to [keep the pace](#), but it doesn't influence anything else. The umas never get blocked or anything. It's just Speed vs Stamina.

Sprint (1400m, Recommendation: 500 Stamina)

1200 Speed 450 Stamina **loses to** 1010 Speed 500 Stamina

1200 Speed 350 Stamina **loses to** 470 Speed 500 Stamina

1200 Speed 250 Stamina **loses to** 130 Speed 500 Stamina

Mile (1800m, Recommendation: 800 Stamina)

1200 Speed 700 Stamina **loses to** 920 Speed 800 Stamina

1200 Speed 600 Stamina **loses to** 560 Speed 800 Stamina

1200 Speed 500 Stamina **loses to** 270 Speed 800 Stamina

Medium (2400m, Recommendation: 900 Stamina + Gold Recovery)

1200 Speed 1000 Stamina **loses to** 710 Speed 1200 Stamina

1200 Speed 900 Stamina **loses to** 470 Speed 900 Stamina + Gold (when it activates)

1200 Speed 600 Stamina + Gold **loses to** 210 Speed 900 Stamina + Gold

Overall with these three, we can roughly say that being 300 Stamina below the recommendation turns your Speed stat into a decoration when compared with healthy umas, and if you're 100 under, then each point of Stamina is worth three points of Speed. Going above the Stamina recommendation will do very little, except insulate you against Debuffers and Rushed, but reaching it is **very** important. On the flip side, this shows how strong Debuffers are. Reducing the enemy's Stamina by 100 is like reducing their Speed by 300, and Nice Nature can reduce it by like 200~400.

Long (3600m, just for fun. Very difficult to have too much Stamina here)

1200 Speed 1000 Stamina **loses to** 200 Speed 1200 Stamina

1200 Speed 1700 Stamina + 3 Golds **loses to** 1000 Speed 1700 Stamina + 4 Golds

1200 Speed 1200 Stamina + 3 Golds **loses to** 700 Speed 1200 Stamina + 4 Golds

1200 Speed 1150 Stamina + 2 Golds **loses to** 200 Speed 1200 Stamina + 4 Golds

Aptitudes



These letters mean different things depending on what row they're in. An A is the baseline and gives no benefit or penalty. If you're below A, you will be penalized, while if you're at an S, you'll receive a small bonus.

The first row is for the surface type, either Turf or Dirt. This rank affects your acceleration. If you have an S, you'll receive a 5% boost. B will give -10%.

The second row is for distance. This rank affects your speed. The rates are the same as the surface type. If you have an S, you'll receive a 5% boost. B will give -10%. If you have an E or lower, it will also reduce your acceleration by a significant amount, but you shouldn't put an uma through that.

The third row is for style. It affects your Wit. An S is a 10% boost, while a B gives -15%. However, this does not affect your chance of triggering skills, only all the other things Wit does. Getting S in style is fairly minor but does increase your team score in Team Trials.

To be precise, here are the exact numbers for each rank.

Type	S	A	B	C	D	E	F	G
Surface	+5%	0%	-10%	-20%	-30%	-50%	-70%	-90%
Dist	+5%	0%	-10%	-20%	-40%	-60%	-80%	-90%
Style	+10%	0%	-15%	-25%	-40%	-60%	-80%	-90%

The +5% to your raw speed or acceleration converts to +10.25% Speed/Power stat (about 120 at 1200 Speed/Power). A -10% would convert to -19% Speed/Power (about -230 at 1200 Speed/Power).

Support Cards In Detail

Stat Stick Passives

Support cards will either be used for their skills or as stat sticks. Comparing skills is easy, but comparing their ability to be a stat stick is more complicated.

Let's look at what's widely considered the best support card in the game, [Kitasan Black](#), at max limit break.

Typically the passives you'll be interested in for a stat stick are "Friendship Bonus," "Training Effectiveness," "Mood Effect," "<stat> Bonus" and "Specialty Priority." "Initial Friendship Gauge" is also convenient as it reduces the time taken to get them to orange bond.

The **Mood Effect** is added to the bonus you get from your current mood, which is +/- 10% per deviation from normal. So, with max mood, it's +20%. With Kitanan's +30% Mood Effect, this gets to +26%. This also applies negatively, so if you're at bottom mood, it would go from -20% to -26%.

Training Effectiveness and **Friendship Bonus** work multiplicatively with this. For unique skills, mood effect and training effectiveness are additive, while friendship bonus is multiplicative. She has a 5% training effectiveness from her unique skill, 10% training effectiveness, and 25% friendship effectiveness. All together, the stat gains will be multiplied by $1.26 * 1.15 * 1.25$ for a total of +81% stats from these passives. Note that friendship and unique friendship will act multiplicatively with each other, not additively. A card with 25% Friendship Bonus and 10% unique friendship bonus will give you 37.5%, not 35%.

Specialty Priority increases how often the card will land on its preferred training. Specialty adds to the weight, while specialty from the uniques multiplies it as if a percent, the same way Friendship Bonus works. Each stat has a base weight of 100, and not appearing has a weight of 50. These are divided by the total to give the chance of appearing on each training, so $100/550 = 18.18\%$ rainbow rate with 0 specialty. For Kitanan, she has 80 base, so $100 + 80 = 180$, then she has unique specialty, so $180 * 1.2 = 216$, which is the same as having 116 base. Her Speed appearance rate would be $216/666$, or 32.4%. (Theoretically, slightly lower due to the chance of the training being full?)

<Stat> Bonus effects, like Kitanan's Power Bonus, add 1 to how much the [training will give you](#) of that stat, before all the modifiers. Kitanan's is Power rather than Speed, which means your big speed rainbows will also give a lot of power. This will also apply when she appears on Power or Guts trainings.



Other Passives

Initial <Stat> is somewhat self-explanatory. At the start of training, the number is added to that stat. This can save you a day or two of training and is particularly useful on cards of types that you're only bringing 1 or 2 of, for example the Power cards in a 4 Speed 2 Power deck.

Hint Frequency - increases the chance of there being hints for the support card (the !s that appear on trainings). Since a hint gives +5 bond, this can help you get rainbows quicker, in addition to making it easier to access the card's skills.

Hint Levels - increases the discount on the skills the hints give you, saving you some skill points if you buy them.

Wit Friendship Recovery - a passive found on Wit (and later Group) cards. When there's a rainbow on the Wit training, it will restore extra energy according to this stat (4 for a MLB SR, 5 for a MLB SSR, except Ikuno SR has none).

Race Bonus - increases the amount of stats and skill points you gain from finishing a race. This includes the +3 all stats from mandatory races and the +10 all stats from URA. No rounding, if you want to go from +3 all to +4 all you need 34% Race Bonus. On most umas, going from 33% to 34% would be +10 all stats.

Fan Bonus - is similar, but applies to the fans gained. This is sometimes relevant for umas with early fan checkpoints, but can usually be ignored.

Pal Card Passives

Note that multiple Pal cards can't appear on the same training, so none of these can stack.

Failure Protection - reduces the chance of failing a training. With a 35% value, a 9% fail chance would go to 6%.

Energy Cost Reduction - reduces the energy that the training the card is on costs. A level 5 Speed training costs 25 energy, but with a 20% reduction, it would only be 20 energy.

Event Recovery - increases the energy you gain from the card's events, most notably the dates.

Event Effectiveness - increases the other rewards, such as the number of skill points gained.

Gold Skill Rates

Some support cards only have a chance to give you a gold skill at the end of their chain, while others are guaranteed to. For example, Fine Motion can either give you Speed Star, or the white version of it. Us JP players called getting the white skill "[agemasen](#)" ("will not give") for short, based on the anime scene where Special Week refuses to give McQueen a thing. "Card didn't give me the gold" is a bit of a mouthful. The chance of it happening is dependent on how high your stat is of the card's type. So, for Fine Motion, it would be based on your Wit. Since the final chains happen sort of randomly, you can't really control this, but it'll let you know if you're justified in complaining at least?

Stat	<400	400+	600+	700+	800+	1000+
Gold Skill Chance	30%	60%	65%	75%	80%	90%

Race Bonus Benefits

The "Race Bonus" passive on support cards increases the stats and skill points you receive from finishing races. This applies to both optional races and the mandatory races and URA Finals. Here's a chart of how many stats and skill points you get from a run based on your total Race Bonus.

Bonus	0%	5%	10%	15%	20%	25%	30%	35%	40%
Stats	351	351	372	372	393	393	414	461	482
SP	870	906	946	991	1044	1080	1123	1165	1218

There's a big jump between 30% and 35% race bonus. In actuality, this jump happens at 34%. If you can pick your supports in a way that gets you to $\geq 34\%$ race bonus, you'll get a good chunk of extra stats. You can use this as a way to break ties between cards that are otherwise similar.

Calculating Training Stat Gain

Unity Cup Scenario		Level				
		1	2	3	4	5
Speed	Speed	8	9	10	11	12
-19 NRG	Power	4	4	4	5	6
Stamina	Stamina	7	8	9	10	11
-17 NRG	Guts	3	3	3	4	5
Power	Power	6	7	8	9	10
-18 NRG	Stamina	4	4	4	5	6
Guts	Guts	6	7	8	9	10
-20 NRG	Speed	3	3	3	4	4
	Power	3	3	3	3	4
Wit	Wit	6	7	8	9	10
+5 NRG	Speed	2	2	2	3	4

URA Scenario		Level				
		1	2	3	4	5
Speed	Speed	10	11	12	13	14
-21 NRG	Power	5	5	5	6	7
Stamina	Stamina	9	10	11	12	13
-19 NRG	Guts	4	4	4	5	6
Power	Power	8	9	10	11	12
-20 NRG	Stamina	5	5	5	6	7
Guts	Guts	8	9	10	11	12
-22 NRG	Speed	4	4	4	5	5
	Power	4	4	4	4	5
Wit	Wit	9	10	11	12	13
+5 NRG	Speed	2	2	2	3	4

These charts give the base values for each training at each level. The energy cost also goes up by 1 at levels 2 and 3, and by 2 at levels 4 and 5. MANT uses the Unity Cup table.

To get the result from training, it's...

$(\text{BaseValue} + \text{Sum of StatBonus}) * (1 + \text{MoodMultiplier} * (\text{Sum of MoodEffect})) * (\text{Sum of TrainingBonus}) * (\text{Product of FriendshipBonus}) * (1 + 0.05 * \text{NumberOfSupportCards})$

Friendship Bonus will only be applied if the support is at orange or higher bond (aka rainbowing).

The MoodMultiplier is 0 at neutral mood, and goes up or down by 0.1 per mood up/down.

Skills

Unity Skills

The skills you get from the Unity scenario have effectiveness based on your team's total stats. Unless you're in a mode where you race alone or low in the Stadium rankings, you will usually have three umas on your team. The skill you get is picked from your team's highest ranked stats, but you can also inherit them. And, yes, they do work outside of the Unity scenario.

Green/Purple skills don't count towards the stat total, but motivation does.

Total Stat	<1200	<1800	<2600	<3600	>=3600
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Average Stat	<400	<600	<867	<1200	>=1200
Multiplier	0.8	0.9	1.0	1.1	1.2

Unique Skills

Unique Skills are unique to each uma. If the uma is 3 star or better, she can pass them down through inheritance. However, unique skills passed down this way are much worse, and their price prevents them from being an appealing option. They're essentially white skills with weird conditions and longer durations.

There are a few that retain some strength and are stronger than a normal white skill with a longer duration, so can be worth taking. These are Oguri Cap's **Triumphant Pulse**(勝利の鼓動), Symboli Rudolf's **Behold Thine Emperor's Divine Might**(汝、皇帝の神威を見よ), Tokai Teio's **Sky-High Teio Step**(究極テイオーステップ), King Halo's **Prideful King(Pride of KING)**, and Fuji Kiseki's **Glittering Star Vaudeville**(煌星のヴォードヴィル).

The level of a unique skill increases its strength. For recovery skills, it's increased by 2% per level. So, a level 4 Unique recovery would restore 5.83% HP instead of 5.5%. For other skills, it's increased by 1% on the first level up, then 3% on each subsequent level up.

Calculating Effectiveness of Skills

Speed Skills

Speed is simple to calculate. Distance is speed multiplied by time, so we can multiply the skill's effectiveness by the duration. Note that the duration is multiplied by track length / 1000, so on longer tracks, the effectiveness will go up.

ハヤテ文字 / **Beeline Burst** - 0.35 over 0.9s

$$0.35\text{m/s} * 0.9\text{s} = 0.315\text{m}$$

$$\text{In } 2400\text{m}, * 2.4 = 0.756\text{m}$$

全身全霊 / **In Body and Mind** - 0.35 over 1.8s

$$0.35\text{m/s} * 1.8\text{s} = 0.63\text{m}$$

$$\text{In } 2400\text{m}, * 2.4 = 1.512\text{m}$$

中距離コーナー◎ / **Medium Corners**◎ - 0.25 over 3s

$$0.25\text{m/s} * 3\text{s} = 0.75\text{m}$$

$$\text{In } 2400\text{m}, * 2.4 = 1.8\text{m}$$

Acceleration Skills

Acceleration is more complicated. You have to calculate how long you'd normally take to accelerate to top speed, as well as how much distance you cover in that time, then compare that to the numbers when accelerating. For these examples we'll assume 1200 Speed and 1000 Power, which is fairly realistic.

With 1000 Power, the acceleration is 0.424m/s/s, ignoring Phase Multipliers..

Time to get to top speed is (Top Speed - Starting Speed) / Acceleration.

Distance is $0.5 * \text{Time} * (\text{Top Speed} + \text{Starting Speed})$

Speed is Starting Speed + Acceleration * Time

地固め / **Groundwork** - 0.2 over 3s

We'll use 17m/s as the starting speed and 20.3m/s as the top speed, which is about what a Front Runner has in Mile.

Without Groundwork, the time taken is $(20.3 - 17) / 0.424 = 7.78\text{s}$

The distance covered is $0.5 * 7.78 * (20.3 + 17) = 145.1\text{m}$

With Groundwork, the time taken is $(20.3 - 17) / 0.624 = 5.29\text{s}$

The distance covered while accelerating is $0.5 * 5.29 * (20.3 + 17) = 98.66\text{m}$

The time spent at top speed is $7.78 - 5.29 = 2.49$, which when multiplied by 20.3 = 50.55m

$98.66 + 50.55 = 149.21$. Subtract the other number, 145.1, and we get **4.11m** gained.

アングリング×スキージング / **Angling and Scheming**, Seiun unique - 0.4 over 4s

This is similar to the Groundwork calculation, but the base speed will be 20m/s, and the top speed will be 24.2m/s, normal numbers for a Front Runner in Mile.

Base Time = $(24.2 - 20) / 0.424 = 9.9\text{s}$

Distance = $0.5 * 9.9 * (24.2 + 20) = 218.8\text{m}$

Angling Time = $(24.2 - 20) / 0.842 = 4.99\text{s}$

Distance = $0.5 * 4.99 * (24.2 + 20) = 110.28\text{m}$

Top Speed Time = $9.9 - 4.99 = 4.91$, times 24.2 = 118.82m

Total = 229.1m, difference = **10.3m**

先手必勝 / **Taking the Lead** - 0.4 over 1.2s

This skill triggers 5 seconds into the acceleration, so we need three steps.

Base Time = $(20.3 - 17) / 0.424 = 7.78\text{s}$

Distance = $0.5 * 7.78 * (20.3 + 17) = 145.1\text{m}$

Speed after 5s = $17 + 0.424 * 5 = 19.12\text{m/s}$

Distance after 5s = $0.5 * 5 * (19.12 + 17) = 90.3\text{m}$

Time With Skill = $(20.3 - 19.12) / 0.824 = 1.43\text{s}$

Distance With Skill = $0.5 * 1.43 * (20.3 + 19.12) = 28.19\text{m}$
 Top Speed Time = $7.78 - (5 + 1.43) = 1.35\text{s}$, times $20.3 = 27.4\text{m}$
 Total = 145.89m , difference = **0.79m**

迫る影 / **Encroaching Shadow** - 0.4 over 0.9s

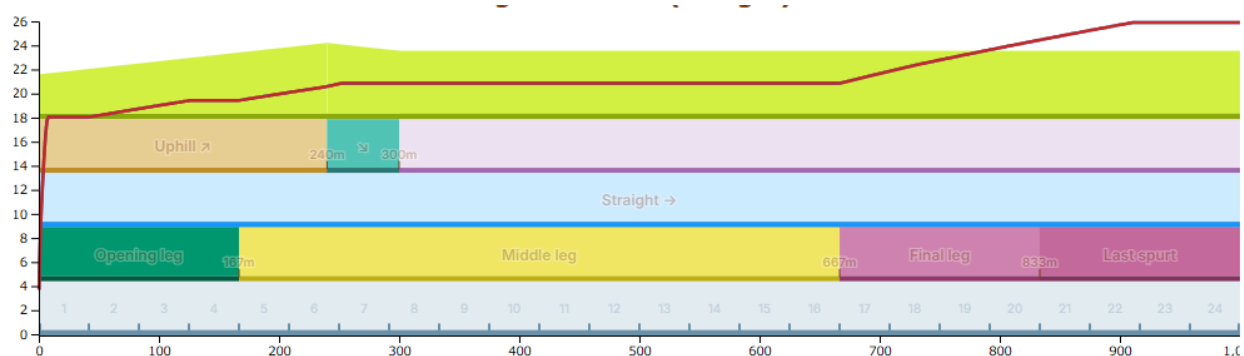
This skill doesn't last long enough, so we have to do three steps again. Taishin gets this skill, and she's used in Medium, so we'll assume 2000m , making the duration 1.8s . As an End Closer, we'll use 19.9m/s as the base speed, and 24.4m/s as the top speed.

Base Time = $(24.4 - 19.9) / 0.424 = 10.61\text{s}$
 Distance = $0.5 * 10.61 * (24.4 + 19.9) = 235\text{m}$

Speed after 1.8s with skill = $19.9 + 0.824 * 1.8 = 21.38\text{m/s}$
 Distance after 1.8s = $0.5 * 1.8 * (21.38 + 19.9) = 37.15\text{m}$
 Time after Skill = $(24.4 - 21.38) / 0.824 = 7.12\text{s}$
 Distance after Skill = $0.5 * 7.12 * (24.4 + 21.38) = 162.98\text{m}$
 Top Speed Time = $10.61 - (7.12 + 1.8) = 1.69\text{s}$, times $24.4 = 41.24\text{m}$
 Total = 241.37m , difference = **6.37m**

Races

How do they work?



Here's an example image of a simple track from the [simulator](#), of a Late Surgeer uma in the Ibis Summer Dash race. The red line is the uma's current speed.

There are four phases of a race, but for the most part we can combine the last two. These phases are: the **Opening Leg**, which lasts for the first 1/6th; the **Middle Leg**, which lasts for the next 3/6ths; the **Final Leg**, which is the next 1/6th; the **Last Spurt**, which is the final 1/6th. Skills refer to these phases, for example, On Your Left! triggers in the Final Leg, and In Body and Mind triggers in the Last Spurt.

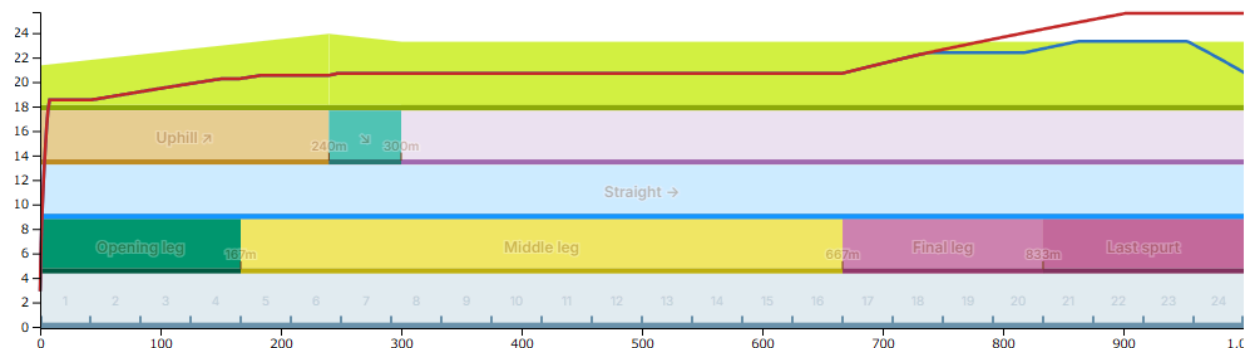
Each uma has a target speed during each phase. For the formulas, check the Technical Doc linked below. For the most part, you can simply know that **umas will run faster in each phase**, except for Front Runners who run slower in the Middle Leg than the Opening Leg. The Final Leg and Last Spurt use the same numbers, so they can be considered as one phase.

Before the Final Leg, your Speed stat **doesn't** affect your target speed. Only Wit affects it slightly. In the Final Leg, after your uma begins her spurt, both Speed and Guts affect your speed. The spurt is when your uma starts spending all her energy to try to finish the race.

When you're not at top speed, you're accelerating. Power affects how fast you accelerate. Umas start out stationary at the start of the race, then accelerate to their Opening Leg speed. They then accelerate to the Middle Leg speed once that phase starts, and then to their Final Leg speed when that starts. If your uma doesn't start her spurt immediately when the Final Leg starts, she'll later accelerate to her spurt speed. During these periods, Acceleration skills are useful, but outside of them, Acceleration skills are useless. The reverse is also true. Speed skills are useless while accelerating, though they can last beyond the acceleration, and if they last into the accel, then your starting point is raised, saving time. (see [Spurt Speed Carry-over](#))

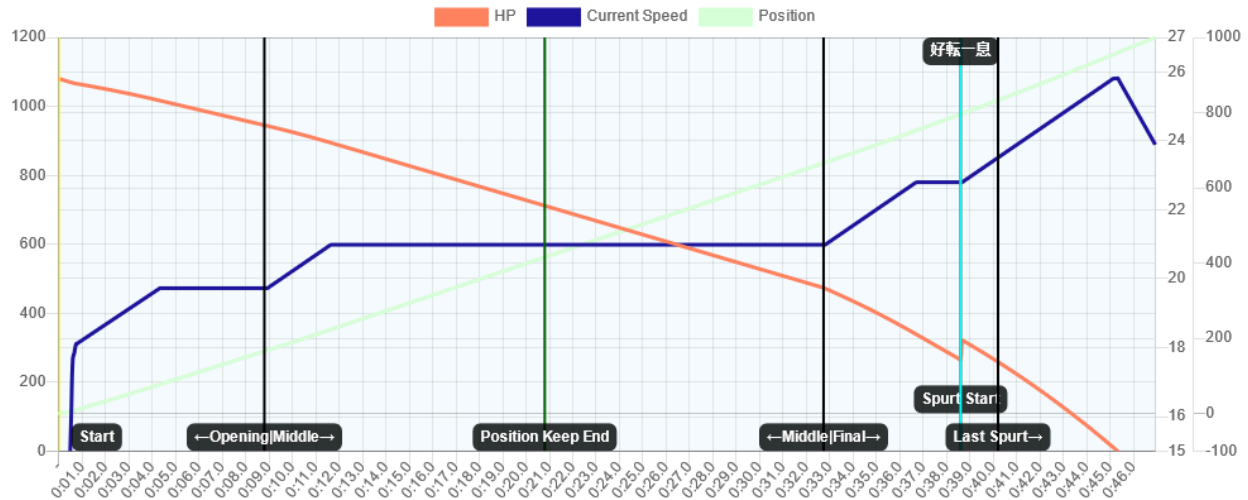
Your uma's speed in the final leg is affected by her remaining HP. When the Final Leg starts, she will decide how fast to run and when to start running fast based on that HP, with higher Wisdom making her do more optimal decisions.

Let's see a chart with a late spurt to see the implications. Red will be the healthy uma, and the blue will be an uma without enough Stamina.



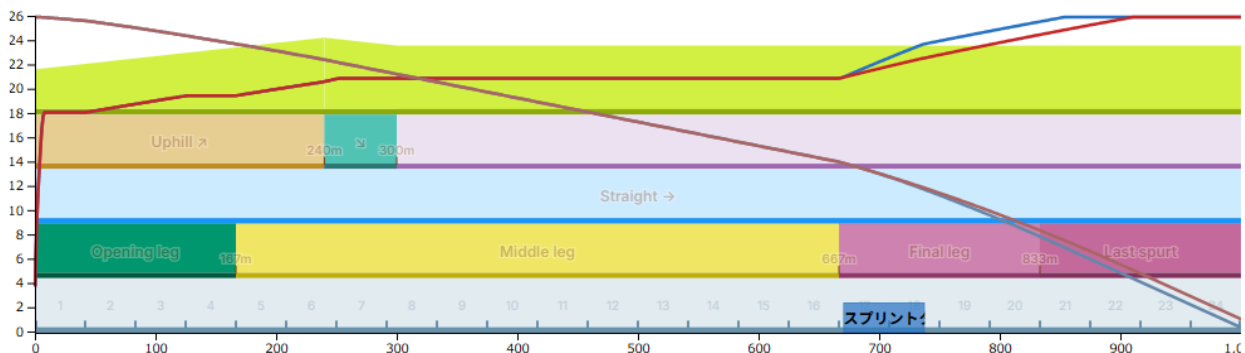
This uma lacks the required Stamina for the track, so she had to delay her spurt. The dip in the speed at the end is what happens when an uma runs out of HP: their speed quickly drops. If your uma is severely lacking in HP, she might not spurt at all, making your Speed stat useless.

If an uma uses a recovery during the last two phases, she'll reconsider her strategy and might alter it. Here's an example. (The blue recovery line is covering the red spurt start line)



In the above graph, the uma didn't have enough HP remaining to start her spurt immediately. However, the Breath of Fresh Air recovery triggered, restoring 5.5% of her HP. With the newfound HP, she decided to start her spurt right then. This isn't as optimal as having the recovery trigger earlier. As you can see, there's a plateau before the recovery, where she wasn't accelerating. If the recovery had triggered earlier, she would have been accelerating all the time, like in the first graph. But, it's still better than in the second graph.

To show the impact of acceleration skills, let's look at a well-timed Turbo Sprint.



This increased acceleration resulted in the blue uma winning by 3.5 lengths. Acceleration skills are very powerful when they work and the meta tends to revolve around them for this reason.

There are some other features of tracks.

Corners: If you remember your high school math classes, the circumference of a circle is $\pi * 2 * \text{radius}$. That means that, the closer you are to the inside of the corner, the less distance you have to run. Umas will naturally move inside for this reason. As a consequence, it's harder to pass someone on a corner, as you have to run a bit more distance than them by going around. Note that each capsule-shaped track has four corners, not two.

Uphills: It's harder to run uphill. Umas will slow down when running uphill, though the higher their Power, the less they slow down. Power becomes more important if there are many uphill.

Downhills: It's easier to run downhill, but only if you know how. While running downhill, umas have a chance, based on their Wit, to get a speed boost and use less HP. On tracks with long downhills, this makes it so high Wit umas have lower Stamina requirements, as well as getting a lot of bonus speed. Wit becomes more important if there are many downhills.

Final Corner: The final corner is what it says on the tin. It's the last corner of the race. On the final corner, the umas will begin to spread out, which helps avoid blocking in the spurt.

Final Straight: The final straight is the straight that comes after the final corner. There must be a corner for there to be a final straight. On Ibis Summer Dash, the track we were looking at, there is no corner, and therefore, no final straight.

Gate Blocks

There are eight gate blocks in a race that the umas start in. If there are more than eight umas, then multiple umas will go into some blocks, starting from the outside. If there are less than eight umas, then the later blocks don't exist.

For example, in CM where there are 9 umas, there are two umas in block 8. In the biggest races, where there are 18 umas, there are three umas in blocks 7 and 8 and two in the rest. Team Trials has twelve umas, so there are two in blocks 5~8.

This adjusts the activation chance of some skills. For example, one might think that Lucky Seven is a $1/8$ chance, but in CM, it's $1/9$, and in the biggest races, it's $3/18$ or $1/6$. Outer Gate benefits from this quite a bit, being $4/9$ in CM and $8/18$ in large races, while Inner Gate is $3/9$ and $6/18$ respectively.

In Team Trials, Lucky Seven is $2/12$, Outer Gate is $6/12$, and Inner Gate is $3/12$.

Though, note that Lucky Seven is only a 50% chance to do something even if you are in gate seven.

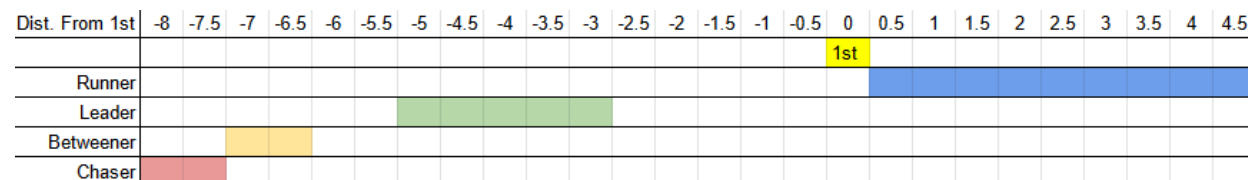
Uma Racing Styles

There are five different styles in the game, if we count Runaway. These mostly dictate what parts of the race your uma tries the hardest in. Here's a quick overview of their performance in each part:

Style	Opening Leg ■□□□□□	Middle Leg □■□□□□	Final Leg □□□□■
Runaway	★★★★★	★★☆☆☆	★★☆☆☆
Front Runner	★★★★☆	★★★★☆	★★☆☆☆
Pace Chaser	★★★☆☆	★★★★☆	★★★☆☆
Late Surger	★★☆☆☆	★★★★☆	★★★★★
End Closer	★☆☆☆☆	★★★★★	★★★★★

For exact numbers, see the Phase Multipliers section below.

During the first half of the race, the umas participate in a dark ritual known as "Position Keep." Each style has a position they want to maintain relative to the furthest-ahead Front Runner. If they fall out of that position, they'll speed up to return to it (Pace Up mode), and if they get too far ahead, they'll **slow down** to return to it. (Pace Down mode) For the nitty gritty details, see the [mechanics doc](#).



The numbers are in meters, but the ones lower than -3 will scale based on the track's length.

Since the styles all base their position off the 1st uma, the better the uma in 1st is, the faster everyone else will run, dictating the pace. The better they are, the less likely Pace Down is, and the more Stamina is needed to finish the race. There's one exception:

- No Front Runner in the race, or the Front Runner is too bad to lead. In this case, a random Pace Chaser will ignore Position Keep and start acting like a Front Runner, with everyone else basing their positions off them. If there are no Pace Chasers too, then a Late Surger, etc. They'll use their normal multipliers, so when there's no Front Runner in the race, Pace Chasers become very scary.

If this is hard to imagine, I made a quick video showing how it works, with the most cutting edge graphics available: <https://www.youtube.com/watch?v=co09NS3BkII>

Phases

There are four different phases in a race. The Opening Leg, which covers the first 1/6th, the Middle Leg, which covers the next 3/6ths, the Last Leg, which is the next 1/6th, and the Last Spurt, which is the final 1/6th.

During each phase, there's a multiplier on the uma's speed and acceleration based on her style.

Speed	Opening Leg ■□□□□□	Middle Leg □■□□□□	Final Leg □□□□■□
Runaway	1.063	0.962	0.95
Front Runner	1	0.98	0.962
Pace Chaser	0.978	0.991	0.975
Late Surger	0.938	0.998	0.994
End Closer	0.931	1	1

Acceleration	Opening Leg ■□□□□□	Middle Leg □■□□□□	Final Leg □□□□■□
Runaway	1.17	0.94	0.956
Front Runner	1	1	0.996
Pace Chaser	0.985	1	0.996
Late Surger	0.975	1	1
End Closer	0.945	1	0.997

Secret Stat Checks

There are secret stat checks on some courses. You will get a +5% speed bonus per 300 in said stat. There's a list of these here

https://twitter.com/umamusu_reveng/status/1377628403403272192. These aren't mentioned in game and have never been officially alluded to, so you can only rely on third party sites for the information.

Stat	Speed Gain
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1~300	+5% Speed stat
301~600	+10% Speed stat
601~900	+15% Speed stat
901+	+20% Speed stat

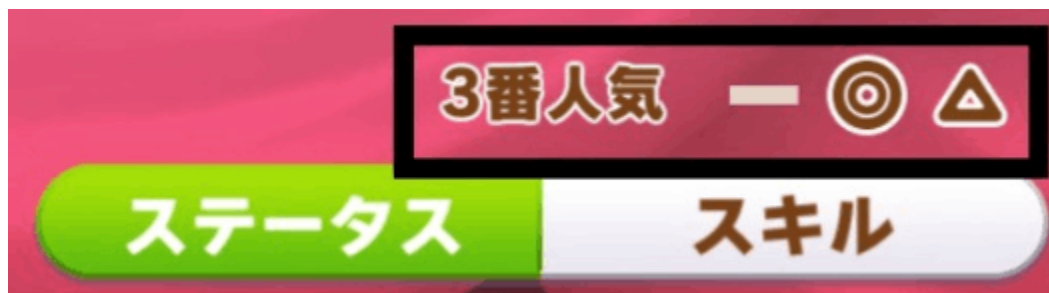
* When there are two, the gain is halved for each

Mood also affects this.

Mood	Stats Needed for Boost		
Highest	290	578	867
Good	296	590	884
Normal	301	601	901
Low	308	614	920
Bottom	314	627	939

Popularity

This is the game's way of evaluating an uma's strength, although not entirely indicative, and is analogous to race betting odds. The basic explanation is that larger relative differences in stats make you more popular, so the more the better if you want a high popularity. The information in this section is based on [this](#).



The symbols from best to worst are ◎, ○, ▲, △. Symbols are only given to the top 5 in evaluations, △ is the only symbol that is shown twice.

Popularity scores are calculated as such.

$$\text{Left symbol} = \phi_{\text{Speed}}(\text{Adjusted speed} \times \phi_{\text{Guts}}(\text{Guts})^{0.5})$$

$$\text{Middle symbol} = \phi_{\text{Stam}}(\text{Adjusted stamina} \times \phi_{\text{Guts}}(\text{Guts})^{0.8} \times \phi_{\text{Wit}}(\text{Wit})^{0.6})$$

$$\text{Right symbol} = \phi_{\text{Power}}(\text{Adjusted power} \times \phi_{\text{Wit}}(\text{Wit})^{0.6})$$

$$\phi_{\text{stat}} = 50 + 10(\text{stat} - \mu_{\text{stat}}) \div \sigma_{\text{stat}}$$

$$\sigma_{\text{stat}} = \text{standard deviation of stat} \quad \mu_{\text{stat}} = \text{mean of stat}$$

Scaled to have a mean of 50 and a standard deviation of 10, aka T-Score (偏差値).

$$\text{Adjusted speed} = (\text{Speed} + \Sigma \text{Adjusted speed skill values}) \times \text{distance aptitude} \times \text{mood}$$

$$\text{Adjusted stam} = (\text{Stam} + \Sigma \text{Adjusted stam skill values}) \times \text{mood}$$

$$\text{Adjusted power} = (\text{Power} + \Sigma \text{Adjusted power skill values}) \times \text{distance aptitude} \times \text{surface aptitude} \times \text{mood}$$

$$\text{Adjusted stat skill value} = \text{Stat value of the skill} \times \text{Activation rate}$$

$$\text{Activation rate for wit check skills} = 1 - 90 \div \text{Wit}$$

$$\text{Activation rate for unique and green skills} = 1$$

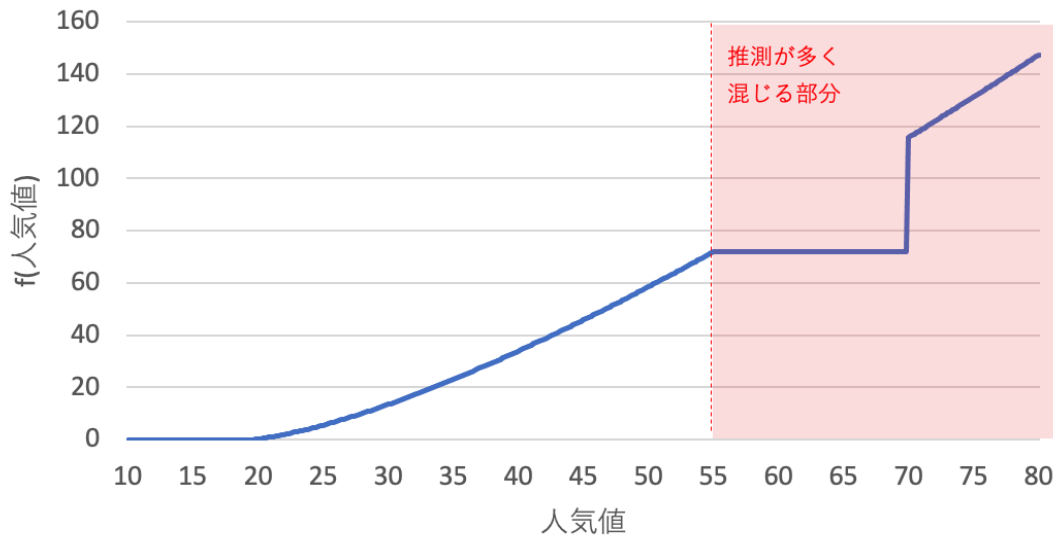
弧線のプロフェッサー	スピード	60		
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For example, Professor of Curvature's adjusted speed skill value at 1200 wit is 55.5.

Guts and wit skill values are not reflected in the evaluations.

How final popularity is calculated concretely is still not fully understood but there are a few more factors to it.

- There is a bonus for having any symbol displayed. The differences between the symbols ◎、○、▲、△ are not that large but it's important that you have one.
- Popularity values over 70 give a large bonus.
- The effect of values that fall between the range of 55 and 70 largely drops off (within this includes a lot of guessing), so your other evaluations that approach 55 or are over 70 will have a greater impact.
- Total popularity is the sum of the function of the three evaluations.



Graph of the function of popularity values.

Manipulating Popularity

The important concepts to understand are averages and deviation. For a high popularity you want stats that are above average, and below for lower popularity. The average stat line of an uma is relative to a single race but you should think about the stats that people want, how much of it they want, and what's realistically achievable to the average player i.e. how much they can train it. There is no clear cut answer but if you assume everyone can achieve the same amount of speed, stamina, and power, then guts and wisdom are the stats you should pay attention to. Guts and wit are used twice for separate calculations in the evaluations as well. Here are my thoughts for each stat.

Speed

Anyone training a real racer will train speed and usually with the goal of reaching the latest scenario's hard cap. Large deviations should be quite impactful since the left symbol is mostly based on speed.

Stamina

The incentives for having extra stamina are a safety net for kakari and the compete before spurt mechanic i.e. avoiding stamina keep. Stamina can vary between hard cap to the minimum amount for a full spurt. For tracks where both stamina and guts decks are viable, stamina has generally been favoured by popularity. Training stamina gives a bit of guts, while training guts gives a bit of speed and power, the same off stats as training speed.

Power

The general goal for all situations has been 1200. Training speed a bunch of times and inheriting power will usually net you around this amount. Achieving a high amount of power in most cases necessitates a sacrifice in at least one other stat either by replacing a would be guts or stamina card slot with a power card.

Guts

Guts is very difficult to raise without directly training guts. Training stamina gives guts but it's merely a byproduct so the difference is usually seen through card strength, stat inherits, or growth bonuses. If you aren't willing to sacrifice wisdom, guts is the next choice.

Wit

The goal is 1200 but most people are willing to sacrifice a bit of it since wit is essentially the RNG stat, which makes it the best stat for manipulating. There isn't a hard number you can pin to wit that will always make a difference in a race such as a difference in speed or power. What matters most is whether your skills activate or not. Basically, you can't see what a 1% difference in wit checks looks like. If most people want 1200, but are willing to settle for less, the relative differences could heavily swing in your favour e.g. the stat difference between 1700 vs 1600 speed and 1300 vs 1200 wisdom are both 100 but wit is proportionally larger.

Technical Information

Refer to this doc, written by KuromiAK, for the nitty gritty details of how racing works:

<https://docs.google.com/document/d/15VzW9W2tXBBTibBRbZ8IVpW6HaMX8H0RP03kq6Az7Xg/edit>